Acoustic communication – the ear, speech and intelligibility

Acoustic communication is what happens when one person speaks and another understands: the sound is produced in the larynx and mouth, sent through a room or a telephone line and analysed by the ear and brain. The course linked physics with psychoacoustics and phonetics. The page has a vowel synthesiser you can listen to and a model of speech intelligibility in rooms.

> DTH, course 5131 (1987), 6 points: Acoustic communication, Acoustics Laboratory. DTH had a strong tradition in the field – from Fritz Ingerslev, who built up the Acoustics Laboratory, to the later Centre for Applied Hearing Research. Denmark today has three of the world's largest hearing-aid manufacturers.

The ear

The outer ear and ear canal collect the sound and amplify it around 3 kHz. The eardrum sets the three bones of the middle ear – hammer, anvil and stirrup – in motion. They act as an impedance transformer: the eardrum is about 17 times larger than the oval window, where the stirrup presses on the fluid-filled inner ear. Without this transformer almost all the sound would be reflected at the transition from air to fluid (see impedance).

In the cochlea a wave travels along the *basilar membrane*. Georg von Békésy showed – and received the Nobel Prize for it in 1961 – that the membrane is stiff and narrow at the base and soft and wide at the apex, so that each frequency gives a maximum at a particular place: high notes near the base, low notes near the apex. The ear is thus a frequency analyser. The hair cells convert the motion into nerve signals, and the outer hair cells actively amplify faint sounds – discovered in 1978, when David Kemp measured sound coming out of the ear: *otoacoustic emissions*, used today to screen the hearing of newborns.

Loudness and masking

Perceived loudness is not proportional to sound pressure. Stevens found that loudness in *sones* roughly doubles for every 10 dB – that is, it grows as the intensity to the power 0.3. The curves of equal loudness (phons) show that the ear is least sensitive at low frequencies and most sensitive around 3–4 kHz.

Fletcher found in 1940 that the ear divides the frequency range into *critical bands*: a tone is masked only by noise within a band of about 100 Hz below 500 Hz and roughly a sixth of an octave above.

Masking spreads mostly upwards in frequency – a low sound hides high sounds better than the reverse. That insight is the basis of MP3 and all modern audio compression: the parts of the signal the ear cannot hear anyway are removed.

The physics of speech

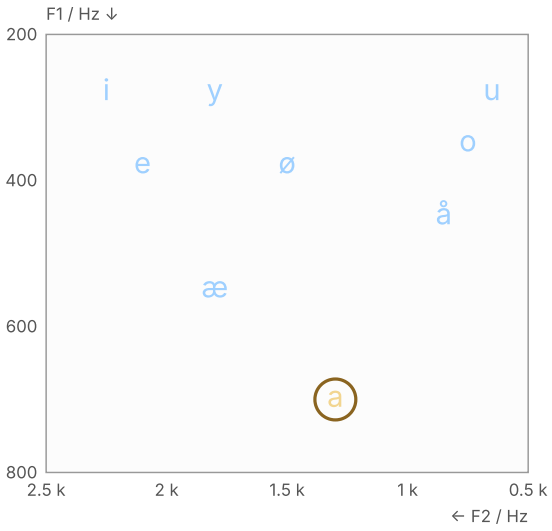
Gunnar Fant's source–filter model of 1960 describes speech as a source that is filtered. The source is either the vibration of the vocal folds – a periodic pulse train with a fundamental frequency of about 120 Hz for men, 220 Hz for women and higher for children, and a spectrum falling by about 12 dB per octave – or turbulent noise at a constriction (s, f, sh). The filter is the pharynx, mouth and nasal cavities, a tube about 17 cm long from vocal folds to lips.

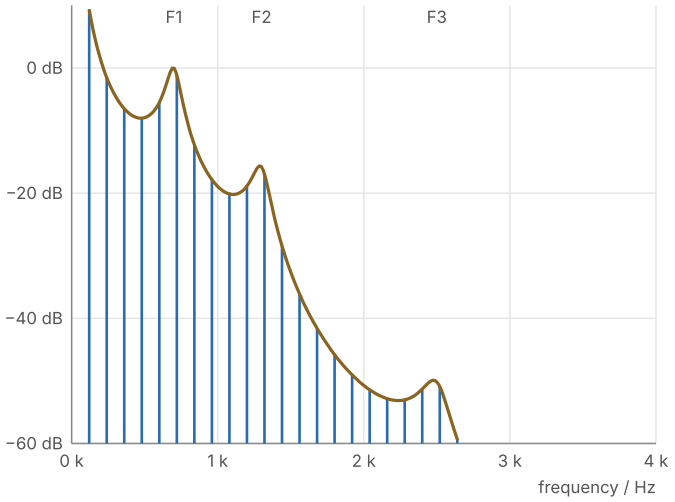
A uniform tube, closed at the vocal folds and open at the lips, has resonances at $f_n = (2n-1)c/4L$ – for $L = 17.5$ cm at 500, 1500 and 2500 Hz. Those are exactly the formants of a neutral vowel (schwa, as in "about"). When the tongue and lips change the shape of the tube, the formants move and we hear different vowels. Most of the energy of speech lies below 1 kHz (the vowels), but most of its intelligibility lies between 1 and 4 kHz (the consonants). That is why the telephone could make do with 300–3400 Hz.

$$f_n = \frac{(2n-1)\,c}{4L} \approx 500,\ 1500,\ 2500\ \text{Hz} \qquad (L = 17.5\ \text{cm})$$

Vowels and formants

The two lowest formants, F1 and F2, almost alone determine which vowel we hear. F1 is related to how open the mouth is (low F1 for i and u, high for a), and F2 to how far forward the tongue is (high F2 for i, low for u and o). Plotting F1 against F2 gives the acoustic vowel quadrilateral, which corresponds to the phoneticians' articulatory one. The figure uses the Danish vowels, including y, ø, æ and å.



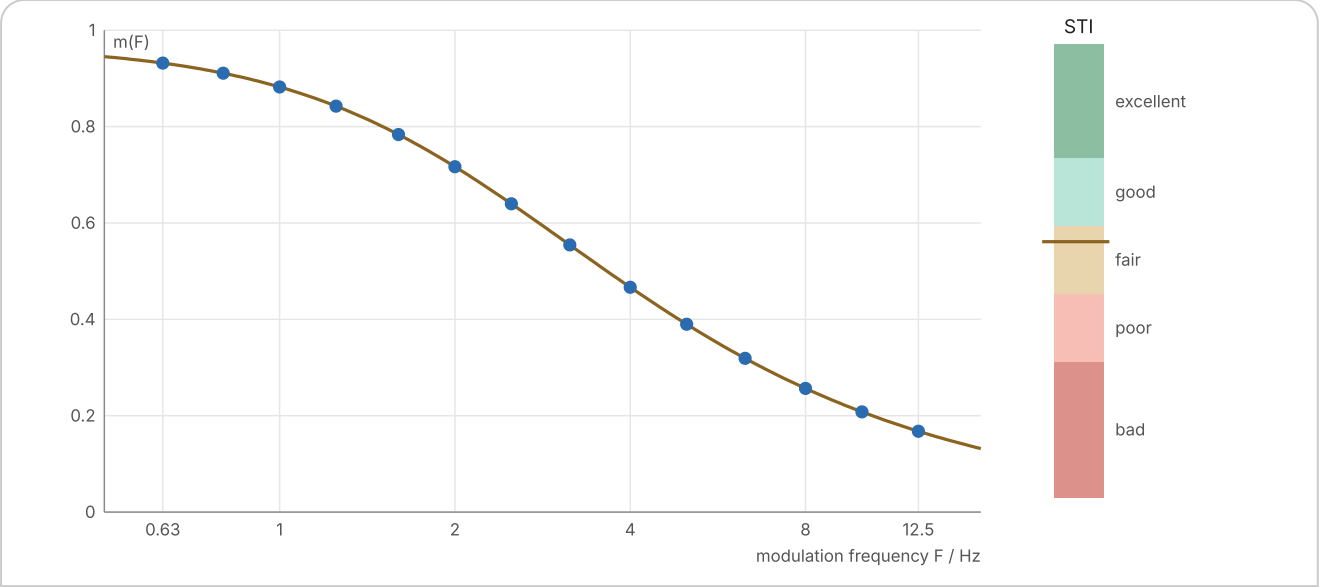


Speech intelligibility

How well speech is understood depends on the signal-to-noise ratio and on the reverberation of the room. Noise fills in the weak parts of speech; reverberation smears out the variations in speech level, so that the gaps between syllables fill with the echo of the previous syllable. In the 1970s Tammo Houtgast and Herman Steeneken introduced the *Speech Transmission Index* (STI), based on exactly this insight: speech is a carrier whose amplitude is modulated at 0.6–12.5 Hz (the syllable rhythm), and one measures how much of the modulation survives transmission. For a room with exponential decay and noise the modulation transfer function is

$$m(F) = \frac{1}{\sqrt{1 + (2\pi F T_{60}/13.8)^2}} \cdot \frac{1}{1 + 10^{-\mathrm{SNR}/10}}$$

Each m is converted into an apparent signal-to-noise ratio $10 \log_{10} \frac{m}{1-m}$, clipped to ±15 dB and scaled to 0–1. The STI is the average. Above 0.6 intelligibility is good, below 0.45 poor. STI is used today to document public-address systems for evacuation in airports and railway stations.



Hearing loss

Almost everyone loses hearing with age (presbycusis), especially at high frequencies, and noise accelerates the loss. When outer hair cells are lost, it is not only sensitivity that disappears but also frequency selectivity: the critical bands widen, masking increases, and it becomes hard to follow a conversation in noise – the "cocktail-party problem" – even when the sound is loud enough. A hearing aid is therefore not just an amplifier: it must amplify differently at different frequencies and levels (dynamic compression), and preferably improve the signal-to-noise ratio with directional microphones.

Since 1989

- **Digital hearing aids** arrived in 1996 – the first almost simultaneously from the Danish companies Oticon and Widex. Today Demant (Oticon), WS Audiology (Widex) and GN Hearing (ReSound) are three of the world's largest manufacturers.
- **Cochlear implants** stimulate the auditory nerve directly with electrodes in the cochlea and have enabled hundreds of thousands of deaf children to hear and speak.
- **Speech recognition:** Where in the 1980s isolated words from one speaker could be recognised, neural networks now transcribe continuous speech from anyone.
- **Audio compression:** MP3 (1993) and AAC are built directly on the masking models of psychoacoustics.

Further reading

- Brian C. J. Moore: *An Introduction to the Psychology of Hearing* (6th ed., Brill 2013)
- Gunnar Fant: *Acoustic Theory of Speech Production* (Mouton 1960)
- Tammo Houtgast and Herman Steeneken: "A review of the MTF concept in room acoustics", *JASA* 77 (1985) 1069–1077

- On this site: Sound synthesiser · Sound analyser

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